

Dashboard & Story Screenshots + Tableau Public Links

Plugging Into the Future: An Exploration of Electricity Consumption Patterns

Team: Dharshini Priya S, Thanusha Peneti, Yogananda Reddy Chinnannagari, Mudiya
Shashank Reddy

1. Dashboard 1 — Monthly Electricity Consumption Overview

Presents the total monthly consumption trend for the selected year alongside a breakdown by region, helping stakeholders spot seasonal peaks at a glance.

S.No	Parameter	Details
4	Calculation Fields Used	Monthly Consumption Total, Average Consumption by Region, Year-over-Year Growth %, Peak Month Indicator
5	Dashboard Design	No. of Visualizations / Graphs — 8: Monthly Consumption Trend, State-wise Consumption Map, Top Consuming States, Consumption by Region, Seasonal Peak Analysis, Year-over-Year Growth, Peak vs. Off-Peak Comparison, Consumption Distribution by Month
6	Story Design	No. of Visualizations / Graphs — 5: National Trend Overview, Regional Comparison, Seasonal Peaks, State Rankings, Year-over-Year Growth

7. Results

7.1 Output Screenshots

Dashboard 1 — "Monthly Electricity Consumption Overview" presents the total monthly consumption trend for the selected year alongside a breakdown by region, allowing users to immediately spot seasonal peaks (typically summer and winter months) and troughs.

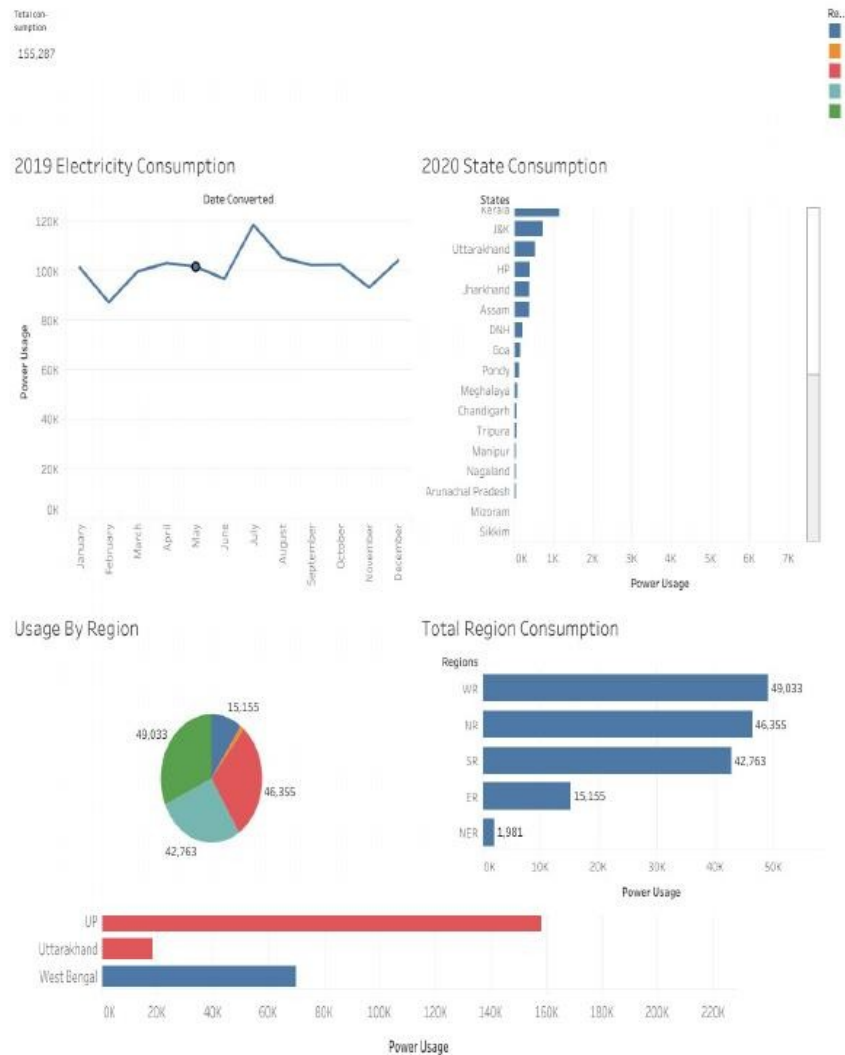
2. Dashboard 2 — State-wise Consumption Comparison

Uses a geographic heatmap and ranked bar chart to show which states/regions consume the most electricity, supporting regional planning and equitable resource allocation.

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Story — "Understanding Electricity Consumption Patterns" walks the viewer through national trends, regional comparisons, seasonal peaks, and year-over-year growth in a guided narrative, making the insights accessible to non-technical stakeholders.

Electricity Consumption Dashboard



3. Story — Understanding Electricity Consumption Patterns

Walks the viewer through national trends, regional comparisons, seasonal peaks, and year-over-year growth in a guided narrative, making the insights accessible to non-technical stakeholders.

Note: this report references the published Tableau workbook "Monthly Electricity Consumption in 2019," part of the Electricity Consumption Analysis project on Tableau Public. Specific screenshots and the underlying processed dataset should be inserted here from the project's GitHub repository and Tableau Public workbook once available for direct embedding.

8. Advantages & Disadvantages

8.1 Advantages

- **Data-Driven Decision Making:** Enables stakeholders to make informed grid and policy decisions based on real-time data insights.
- **Interactive Dashboards:** Tableau dashboards provide dynamic visualizations that allow users to explore consumption data in a user-friendly manner.
- **Improved Demand Understanding:** Helps identify seasonal peaks, regional disparities, and growth trends in electricity usage.
- **Supports Grid Planning:** Insights can guide capacity planning and demand-side management strategies.
- **Time Efficiency:** Reduces the time needed to interpret large consumption datasets by converting them into intuitive visuals.
- **Scalability:** Dashboards can be updated with new data to reflect current trends without reworking the entire project.

8.2 Disadvantages

- **Data Dependency:** The quality of insights is limited by the quality and completeness of the underlying consumption dataset.
- **Tool Limitations:** Tableau Public has feature limitations compared to the paid version (e.g., data refresh, privacy controls).
- **Learning Curve:** New users may require time to understand and navigate Tableau's interface effectively.
- **Limited Predictive Capabilities:** Basic dashboards focus on descriptive analytics; forecasting requires integration with additional tools.
- **Internet Requirement:** Access to dashboards hosted on Tableau Public or cloud services requires a stable internet connection.

9. Conclusion

The project "Plugging into the Future: An Exploration of Electricity Consumption Patterns" demonstrates the powerful role of data visualization in understanding and responding to electricity demand. As energy needs continue to grow and diversify, utilities and policy makers are under increasing pressure to plan capacity, manage seasonal peaks, and support sustainable growth. This project addresses these challenges by transforming raw, complex consumption data into visually appealing and interactive dashboards using Tableau.

4. Tableau Public Links

Published workbook:

https://public.tableau.com/views/Electricity_Consumption_Analysis_17840432823280/MonthlyElectricityConsumptionin2019?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Profile: public.tableau.com/app/profile/dharshini.priya.s